

Retail Store Sales Analysis — Auto Analysis

Business Analytics Report

Project: Retail Store Sales Analysis

Dataset: retail_sales_dataset.csv

Analysis date: 2026-09-25

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Generated by the AI Data Scientist Platform

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1. Executive Summary

This report analyzes **retail_sales_dataset.csv** (3,000 rows, 14 columns) covering 2024-01-02 to 2026-06-30 to answer the business question: *A synthetic dataset of 3,030 retail orders from five Pakistani cities (Jan 2024 to Jun 2026), with 14 columns covering order date, store city, product category, sales channel, customer segment, unit price, quantity, discount, promotion, customer rating, sales amount, profit, and returns. It includes missing values, duplicate rows, quantity outliers, and invalid dates for testing data cleaning. Suitable for sales regression, return prediction, and time-trend analysis.*

Key verified findings

- The original dataset contained 426 missing value(s) and 30 duplicate row(s), which have since been cleaned.
- A Gradient Boosting model was trained to predict **Sales Amount** (regression). On average, predictions are off by about 36.80 units (Mean Absolute Error).
- 1 feature excluded as data leakage before training (Profit) to avoid an artificially inflated score.

Model reliability, in plain terms: On average, predictions are off by about 36.80 units (Mean Absolute Error). The model explains about 98.8% of the variation in the target ($R^2 = 0.988$).

Top recommendations

- 'Unit Price' is the strongest single predictor of Sales Amount in this model (relative importance 50.7%).
- 'Product Category: Electronics' is the second most influential predictor of Sales Amount in this model (relative importance 24.9%).
- 'Quantity' is also a meaningful predictor of Sales Amount in this model (relative importance 23.5%).

2. About the Dataset

retail_sales_dataset.csv contains 3,000 rows and 14 columns. It spans the period 2024-01-02 to 2026-06-30 (based on Order Date).

Business question: A synthetic dataset of 3,030 retail orders from five Pakistani cities (Jan 2024 to Jun 2026), with 14 columns covering order date, store city, product category, sales channel, customer segment, unit price, quantity, discount, promotion, customer rating, sales amount, profit, and returns. It includes missing values, duplicate rows, quantity outliers, and invalid dates for testing data cleaning. Suitable for sales regression, return prediction, and time-trend analysis.

Column dictionary

Column	Type	Missing	Notes
Order Id	Identifier	0.0%	
Order Date	Date/Time	0.0%	
Store City	Category	0.0%	most common: Karachi
Product Category	Category	0.0%	most common: Sports
Sales Channel	Category	0.0%	most common: In-Store
Customer Segment	Category	0.0%	most common: Regular
Unit Price	Numeric	0.0%	typical range 7.21–559.21
Quantity	Numeric	0.0%	typical range 1.00–144.00
Discount Percent	Numeric	0.0%	typical range 0.00–25.00
Promotion Applied	Category	0.0%	most common: No
Customer Rating	Numeric	0.0%	typical range 1.20–5.00
Sales Amount	Numeric	0.0%	typical range 6.82–5,605.46

Profit	Numeric	0.0%	typical range 0.09–1,932.94
Returned	Category	0.0%	most common: No

3. Data Quality & Cleaning

Original dataset quality (as uploaded)

Total rows	Missing cells	Duplicate rows	Issues detected
3,030	426	30	12

Duplicate Rows

Column	Finding	Affected
(whole dataset)	30 duplicate row(s) detected.	30

Invalid Dates

Column	Finding	Affected
order_date	'order_date' has 5 value(s) that don't parse as a valid date.	5

Missing Values

Column	Finding	Affected
sales_channel	'sales_channel' has 61 missing value(s) (2.0%).	61
customer_segment	'customer_segment' has 90 missing value(s) (3.0%).	90
unit_price	'unit_price' has 92 missing value(s) (3.0%).	92
customer_rating	'customer_rating' has 183 missing value(s) (6.0%).	183

Outliers

Column	Finding	Affected
unit_price	'unit_price' has 482 statistical outlier(s) outside [-61.73, 175.88].	482
quantity	'quantity' has 99 statistical outlier(s) outside [-1.5, 10.5].	99
customer_rating	'customer_rating' has 4 statistical outlier(s) outside [1.65, 6.05].	4
sales_amount	'sales_amount' has 400 statistical outlier(s) outside [-396.01, 919.1].	400

profit	'profit' has 362 statistical outlier(s) outside [-120.46, 257.28].	362
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Impossible Values

Column	Finding	Affected
profit	'profit' has 29 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	29

Cleaning performed

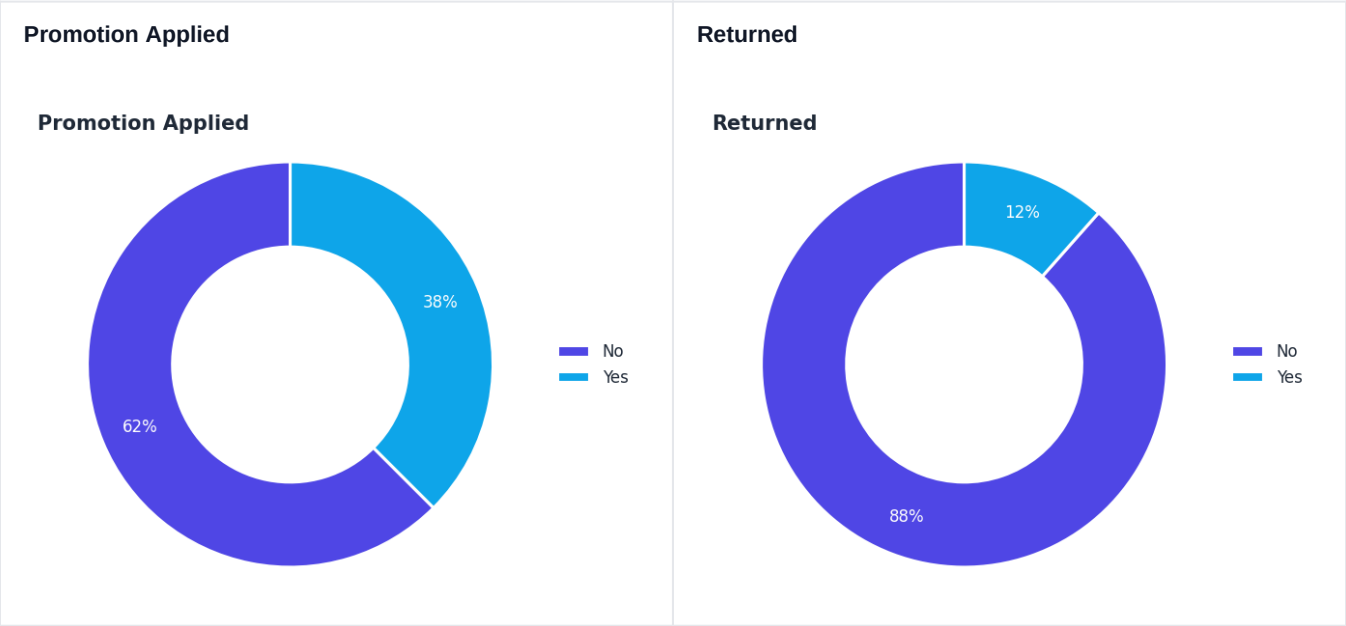
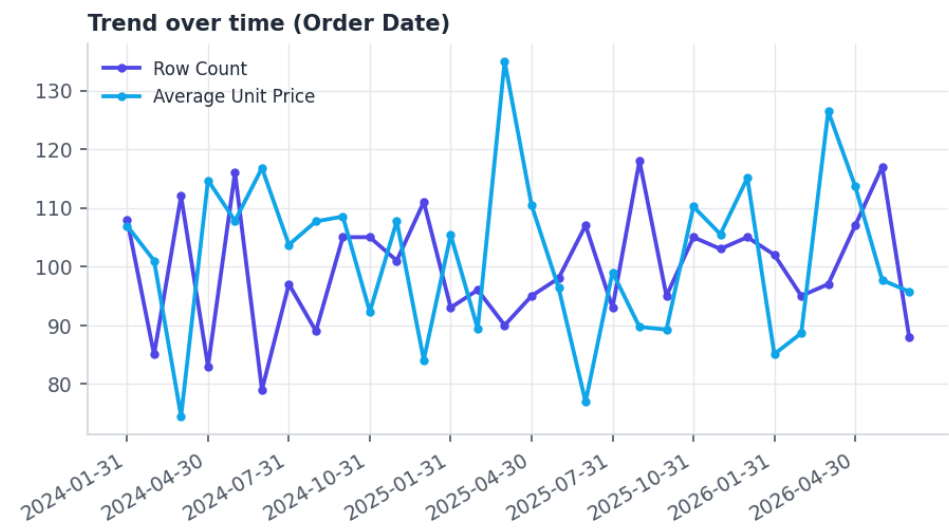
Problem	Before	Action	After	Reason
Duplicate rows	30 duplicate row(s)	Removed duplicate rows	0 duplicate rows	Duplicate records would otherwise double-count the same transaction/customer.
Missing Sales Channel	61 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Customer Segment	90 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Unit Price	92 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Missing Customer Rating	183 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Impossible negative Profit	29 impossible value(s)	Converted to missing, then filled	0 impossible values	A rare negative value in an otherwise non-negative column is almost always a data-entry error.

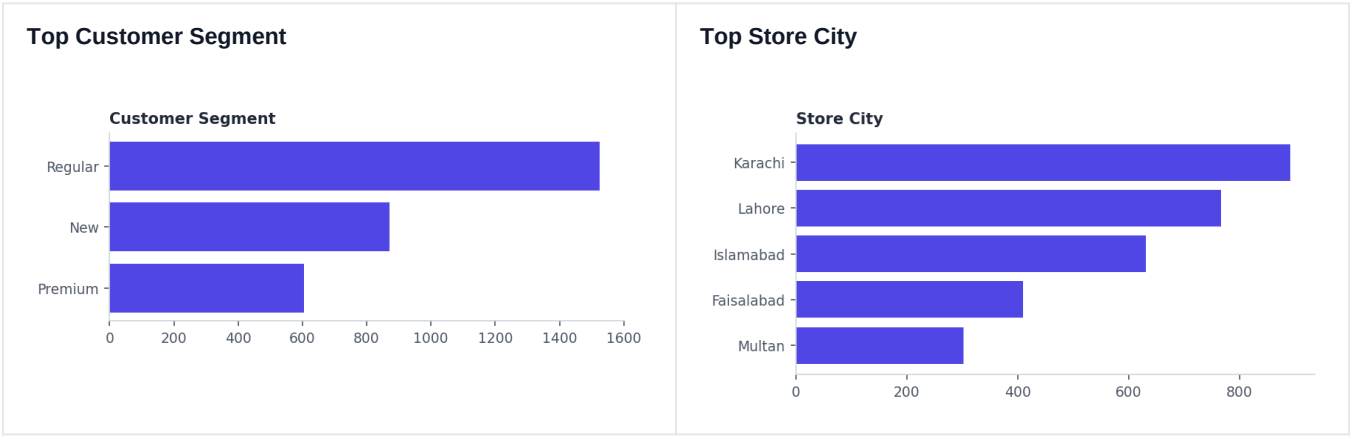
After cleaning: 0 missing cell(s), 0 duplicate row(s), 3,000 rows remain. The original file itself was never modified — cleaning produced a separate working copy.

4. Key EDA Findings

Total Rows	Average Sales Amount	Total Unit Price	Total Quantity
3,000	469	303,813	15,405

Trend over time (Order Date)





Discount Percent breakdown

Average: 7.77 (range 0.0–25.0)

Value	Count	Percent
25	150	5.0%
20	316	10.5%
15	281	9.4%
10	601	20.0%
5	602	20.1%
0	1050	35.0%

5. Modeling Approach

A **regression** model was trained to predict **Sales Amount**, using 13 feature(s).

Features used

Order Date: Year, Order Date: Month, Order Date: Dayofweek, Store City, Product Category, Sales Channel, Customer Segment, Unit Price, Quantity, Discount Percent, Promotion Applied, Customer Rating, Returned

Features excluded, and why

Column	Reason excluded
Order Id	Identifier column (e.g. an ID) — carries no predictive signal.
Order Date	Raw date — replaced with derived year / month / day-of-week features instead.
Profit	Leakage. 'profit' is named like a value computed FROM other monetary fields (e.g. profit = revenue - cost), and correlates with the target here (89%) — likely calculated alongside 'sales_amount' from the same transaction rather than known beforehand, which would let the model 'cheat' by learning a near-reconstruction of the target.

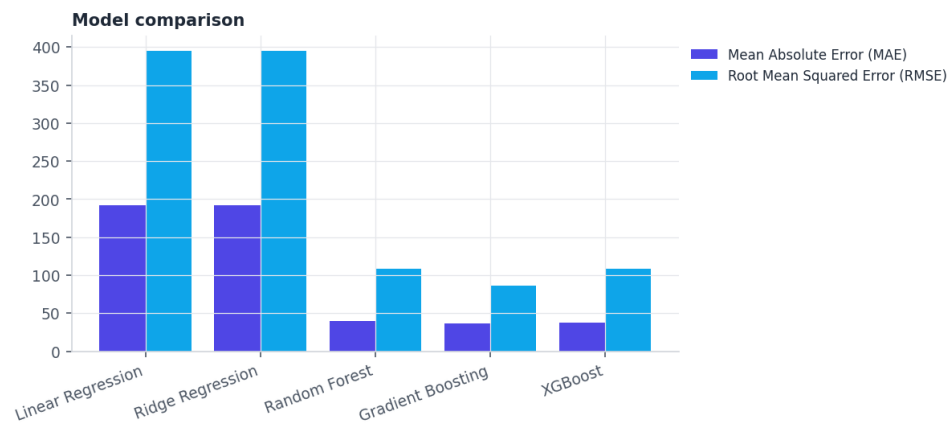
Reviewed, not excluded: Unit Price (85.0% correlated) — no evidence these are derived from the target, so they were kept as legitimate predictors, but worth a manual sanity check.

Technical details
Algorithm: Gradient Boosting · Preprocessing: 7 numeric + 6 categorical feature(s), scaled/encoded · Train/test split: 20.0% held out · Cross-validation folds: 5 · Random seed: 42

6. Model Performance

Model comparison

Model	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R ² (variance explained)
Linear Regression	192.1270	395.5135	0.7424
Ridge Regression	192.0733	395.5292	0.7424
Random Forest	40.0927	108.8667	0.9805
Gradient Boosting	36.7963	86.5221	0.9877
XGBoost	37.7625	108.4177	0.9806

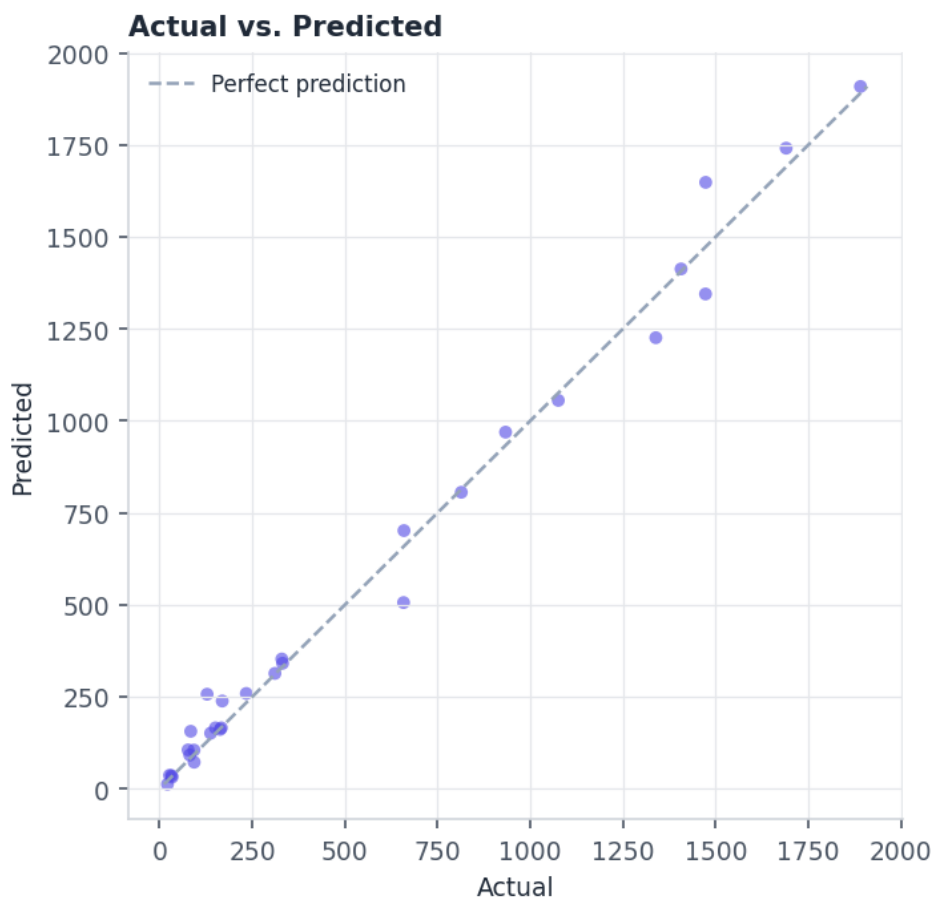


In plain language

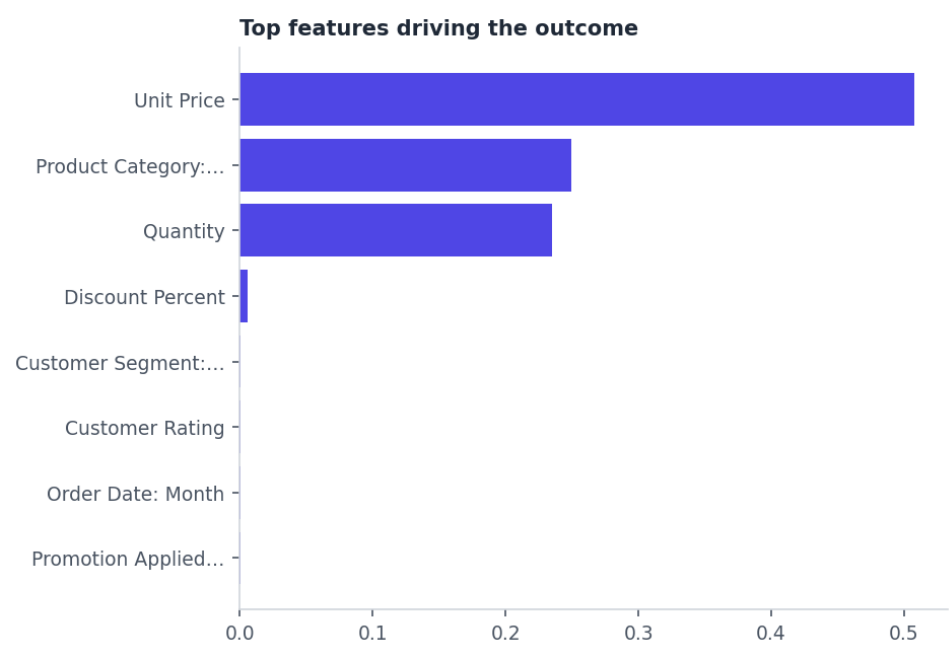
- On average, predictions are off by about 36.80 units (Mean Absolute Error).
- The model explains about 98.8% of the variation in the target ($R^2 = 0.988$).

Compared against a trivial baseline (R^2 (variance explained) = -0.0031): this model clearly outperforms it.

Actual vs. Predicted



7. What Drives the Outcome



Feature	Relative importance
Unit Price	50.7%
Product Category: Electronics	24.9%
Quantity	23.5%
Discount Percent	0.6%
Product Category: Groceries	0.1%
Customer Segment: Regular	0.1%
Customer Rating	0.0%
Order Date: Month	0.0%
Promotion Applied: Yes	0.0%
Promotion Applied: No	0.0%

8. Future Outlook

- 'Unit Price' is the strongest single predictor of Sales Amount in this model (relative importance 50.7%).
- 'Product Category: Electronics' is the second most influential predictor of Sales Amount in this model (relative importance 24.9%).
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Example verified predictions

Actual	Predicted
1406.83	1412.87
21.58	12.49
166.96	165.69
81.68	91.87
93.14	105.19

9. AI Business Insights AI-generated

AI insights are unavailable right now (gemini: Gemini rate limit exceeded — too many requests. Try again shortly.; ollama fallback: The AI service is currently unavailable.). The rest of this report contains verified analytical results only.

10. Limitations

- The following column(s) were excluded as likely data leakage and could not be used: profit. If any of these should genuinely be available before prediction time, review them manually.
- Column(s) with high correlation to the target were retained but should be manually verified as legitimate, known-in-advance predictors: unit_price.
- Cross-validation scores varied by ± 0.008 across folds — treat the headline metric as an estimate with this much natural uncertainty, not an exact figure.
- Model predictions are estimates based on historical patterns and may not hold if underlying business conditions change. They should inform, not replace, domain expertise.

11. Appendix

Full model metrics

Metric	Value
Mean Absolute Error (MAE)	36.7963
Mean Squared Error (MSE)	7486.0786
Root Mean Squared Error (RMSE)	86.5221
R ² (variance explained)	0.9877
Mean Absolute Percentage Error	0.1291
Cv Mean	0.9821
Is Weak	False

Preprocessing

Setting	Value
Test Size	0.2
Cv Folds	5
Random seed	42
Model version	4

Hyperparameters

Parameter	Value
Random Seed	42

Dataset statistics

Statistic	Value
Rows	3,000
Columns	14